

Luca Buiarelli

GitHub | Google Scholar | Website

+1 952 999 6773
+39 331 106 9536
buiar001@umn.edu

Profile

Condensed matter physicist with experience in analytical modeling, numerical simulations, and group theory. Expertise includes Landau theories, tight-binding models, density functional theory, and Monte Carlo methods, with a current research focus on altermagnetism and unconventional magnetic phases. I developed **ProDenCeR**, an open-source Python tool for post-processing density functional theory calculations to diagnose unconventional electronic and magnetic phases in materials.

Education

University of Minnesota, USA — PhD, Materials Science	Sep. 2023 – Present
University of Copenhagen, Denmark — MSc, Condensed Matter Physics	Sep. 2021 – May 2023
University of Pisa, Italy — BSc, Physics	Sep. 2018 – Sep. 2021

Awards and Fellowships

Matthew Tirrell Graduate Fellowship	2024
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Teaching

Teaching Assistant	Spring and Fall 2024
University of Minnesota, Introduction to Materials Science Laboratory (MATS 2001/2002)	

Service and Outreach

Journal Refereeing

Nature *npj Quantum Materials*, Nature *Communications Materials*

Penne Amiche della Scienza

Italy

Science outreach program for Italian middle school students involving monthly letter exchanges, aimed at fostering interest in science and academic careers.

Workshops and Conferences

Aspen Center for Physics, Winter Conference on Altermagnetism, Aspen, CO (poster)	April 2026
American Physical Society March Meeting, Denver, CO (contributed talk)	March 2026
Flatiron Institute, TRIQS School, Paris, France (attended)	September 2025
American Physical Society March Meeting, Los Angeles, CA (contributed talk)	March 2025
School on Electron-Phonon Physics, University of Texas at Austin (attended)	June 2024
American Physical Society March Meeting, Minneapolis, MN (contributed talk)	March 2024

Skills

Coding: MATLAB, Python, C

Scientific Communication: Experience preparing scientific manuscripts using $\text{\LaTeX}$ and delivering oral presentations with PowerPoint slides at international conferences.

Languages: English (fluent), Italian (native), Spanish (basic)

Articles

Published

1. Interplay between charge correlations and superconductivity across the superconducting domes of $\text{CsV}_3\text{Sb}_{5-x}\text{Sn}_x$
 Andrea N. Capa Salinas, Brenden R. Ortiz, Steven J. Gomez Alvarado, Sarah Schwarz, Ganesh Pokharel, **Luca Buiarelli**, Hyeonseo Harry Park, Shiyu Yuan, Roland Yin, Suchismita Sarker, Turan Birol, Stephen D. Wilson
 Phys. Rev. B **114** (2026) [Journal] [arXiv]
2. Probing multipolar order in the candidate altermagnet MnF_2 through the elastocaloric effect under strain
 Rahel Ohlendorf, **Luca Buiarelli**, Hilary ML Noad, Andrew P Mackenzie, Rafael M Fernandes, Turan Birol, Jörg Schmalian, Elena Gati
 Phys. Rev. Lett. **137** (2026) *Editors' Suggestion* [Journal] [arXiv]
3. Observation of mirror-odd and mirror-even spin texture in ultra-thin epitaxially-strained RuO_2 films
 Yichen Zhang, Seung Gyo Jeong, **Luca Buiarelli**, Seungjun Lee, Yucheng Guo, Jiaqin Wen, Hang Li, Sreejith Nair, In Hyeok Choi, Zheng Ren, Ziqin Yue, Alexei Fedorov, Sung-Kwan Mo, Junichiro Kono, Jong Seok Lee, Tony Low, Turan Birol, Rafael M. Fernandes, Milan Radovic, Bharat Jalan, Ming Yi
 Science Adv. **12** (2026) [Journal] [arXiv]
4. Electronic and structural properties of Rh- and Pd-based kagome layered shandites from first principles
Luca Buiarelli, Turan Birol, Brian M. Andersen, Morten H. Christensen
 Phys. Rev. B **113** (2026) *Editors' Suggestion* [Journal] [arXiv]
5. Altermagnetic Polar Metallic Phase in Ultra-Thin Epitaxially-Strained RuO_2 Films
 Seung Gyo Jeong, In Hyeok Choi, Sreejith Nair, **Luca Buiarelli**, Bitu Pourbahari, Jinyoung Oh, Nabil Bassim, Ambrose Seo, Woo Seok Choi, Rafael M. Fernandes, Turan Birol, Liuyan Zhao, Jong Seok Lee, Bharat Jalan
 Proc. Natl. Acad. Sci. **123** (2026)[Journal] [arXiv]
6. Noncollinear Magnetic Multipoles in Collinear Altermagnets
Luca Buiarelli, Rafael M. Fernandes, Turan Birol
 Phys. Rev. B **112**, 224442 (2025) [Journal] [arXiv]

Preprints

1. Discovery of an odd-parity f-wave charge order in a kagome metal
 Jiangchang Zheng, Caiyun Chen, Ruiqin Fu, **Luca Buiarelli**, Zihan Lin, Fazhi Yang, Tianhao Guo, Ganesh Pokharel, Andrea Capa Salinas, Sen Zhou, Turan Birol, Stephen D. Wilson, Junzhang Ma, Daniel J. Schultz, Xianxin Wu, Berthold Jäck
 Submitted [arXiv]